

NBR-RELATED PATENT LANDSCAPE: POLYMERIZATION, COMPOSITION CONTROL, AND PROCESS-DRIVEN RUBBER PERFORMANCE

1. ABSTRACT

This report synthesizes two primary patents that mention nitrile-butadiene rubber (NBR) in distinct technical contexts: EP2507313B1, which focuses on accelerator-free elastomeric glove films based on carboxylated acrylonitrile-butadiene latex, and US10317811B2, which uses NBR as a vulcanized rubber component in an electroconductive charging roller. Although neither patent is a dedicated NBR polymerization patent, both provide useful evidence on NBR formulation, compounding, and downstream processing constraints relevant to NBR-based materials.

EP2507313B1 is the more synthesis-relevant document because it discloses the latex composition architecture, including acrylonitrile/butadiene/carboxylic acid ratios, methacrylic acid premixing, pH adjustment, total solids control, and sulfur/accelerator-free crosslinking using metal oxides. The patent emphasizes self-crosslinked or methacrylic-acid-modified carboxylated nitrile latex for glove dipping, followed by drying and thermal curing, with performance benefits such as reduced sulfur residues and improved aging behavior. US10317811B2 is not a synthesis patent in the classical sense, but it discloses an NBR-containing unvulcanized rubber composition for extrusion molding of a charging roller, followed by air vulcanization and ultraviolet surface treatment. The examples show how NBR is compounded with carbon black, zinc oxide, zinc stearate, calcium carbonate, sulfur, and TBzTD, and how process variables such as take-up rate, vulcanization temperature, and UV dose affect surface hardness, viscosity, and image performance. Across the two patents, the dominant trends are latex-based formulation control, careful management of crosslinking chemistry, and process conditions that tune mechanical properties rather than monomer synthesis itself. For NBR synthesis and manufacturing, the evidence supports the importance of controlling acrylonitrile content, carboxylation or functional modification where needed, and downstream cure strategy to balance elasticity, durability, and surface performance.

2. METHODOLOGY

PRIMARY PATENT EVIDENCE

Patent 1

- Patent Number: EP2507313B1
- Patent Title: Elastomeric rubber and rubber products without the use of vulcanizing ...
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2018
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: The patent discloses an accelerator-free latex-based process for making elastomeric rubber films, especially glove films, using a carboxylated acrylonitrile-butadiene latex that is either premixed with methacrylic acid or used as a self-crosslinked latex. The deposited latex layer is formed on a former and then vulcanized without free sulfur and without vulcanization accelerators, relying on metal oxide-assisted crosslinking and pH-controlled formulation. - pH: 9-10 — Claims - Total solid content (TSC): 18-30% — Claims - Acrylonitrile content: 26-30 wt% — Claims - Butadiene content: 65-58 wt% — Claims - Carboxylic acid content: 6-8 wt% — Claims - Methacrylic acid premix: 2-8 wt% per 100 wt% of carboxylated acrylonitrile-butadiene latex — Claims - Calcium ion concentration: 5.0-20.0% — Example 2 - Calcium ion concentration: 8.0-17.0% — Example 2 - Drying temperature: 50-70 degree C — Example 2 - Dipping time: 1-20 seconds — Example 2 - Preferred dipping time: 12-17 seconds — Example 2 - Drying temperature: 80-120 degree C — Example 2 - Drying time: 20-70 seconds — Example 2 - Resultant solid substances: 30% by weight — Example 3 - Curing temperature: 150°C — Example 6 - Humidity conditioning: 50% — Example 6 - Conditioning temperature: 23°C — Example 6 - Aging temperature: 70 °C — Example 6 - Aging time: 7 days — Example 6 - Tensile strength: 26 MPa — Example 6 - Tensile strength after aging: 34 MPa — Example 6 - Elongation: 620% — Example 6 - Elongation after aging: 640% — Example 6 - Modulus at 500%: 9.8 MPa — Example 6 - Modulus at 500% after aging: 12.6 MPa — Example 6 - Conventional tensile strength: 26 MPa — Example 6 - Conventional tensile strength after aging: 26 MPa — Example 6 - Conventional elongation: 600% — Example 6 - Conventional elongation after aging: 620% — Example 6 - Conventional modulus at 500%: 12.3 MPa — Example 6 - Conventional modulus at 500% after aging: 10.6 MPa — Example 6

Relevant Experimental Evidence - Example 2: Demonstrated glove dipping and drying steps using compounded latex, calcium-ion coagulant, and multi-step drying/dipping conditions. - Example 3: Showed a comparative glove made from self-crosslinked carboxylated nitrile latex with 30 wt% solids and no sulfur/accelerators. - Example 4: Elemental analysis indicated sulfur was non-detectable in the inventive glove and zinc content was reduced relative to the conventional product. - Example 6: Inventive gloves cured at 150°C showed comparable tensile strength and improved aging performance versus sulfur/accelerator-cured gloves.

Technical Relevance This patent is highly relevant because it directly addresses NBR-family latex formulation and crosslinking strategy for practical rubber products. It provides explicit composition ranges and processing conditions that are useful for designing accelerator-free nitrile

latex systems with controlled mechanical performance.

Patent 2

- Patent Number: US10317811B2
- Patent Title: Charging member, method for producing same, process cartridge and ...
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2019
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: The patent discloses production of a charging roller by preparing an unvulcanized rubber composition containing a polymer having a butadiene skeleton, including NBR in the examples, then molding it by crosshead extrusion onto an electroconductive support. The rubber layer is vulcanized in air and subsequently UV-irradiated to create an oxidatively cured topmost surface with roughened insulating-particle exposure. - Carbon black in electroconductive substrate: 10% by mass — Example 1 - Diameter of stainless steel substrate: 6 mm — Example 1 - Length of stainless steel substrate: 252.5 mm — Example 1 - Carbon black: 50 parts by mass — Example 1 - Zinc oxide: 5 parts by mass — Example 1 - Calcium carbonate: 30 parts by mass — Example 1 - Zinc stearate: 1 part by mass — Example 1 - NBR: 100 parts by mass — Example 1 - Kneading time: 15 minutes — Example 1 - Mixer temperature: 50° C. — Example 1 - Capsule particle 1: 5 parts by mass — Example 1 - Sulfur: 1 part by mass — Example 1 - TBzTD: 4 parts by mass — Example 1 - Kneading time: 10 minutes — Example 1 - Double roll machine temperature: 25° C. — Example 1 - Molding temperature: 100° C. — Example 1 - Screw rotation: 9 rpm — Example 1 - Average take-up rate: 107% — Example 1 - Die inner diameter: 8.0 mm — Example 1 - Outer diameter at center: 8.25 mm — Example 1 - Outer diameter at positions 100 mm from center: 8.10 mm — Example 1 - Vulcanization temperature: 160° C. — Example 1 - Vulcanization time: 1 hour — Example 1 - UV wavelength: 254 nm — Example 1 - Integrated UV amount: 9000 mJ/cm² — Example 1 - Grinding grain size: 100 mesh — Comparative Example 6 - Roller rotations: 400 rpm — Comparative Example 6 - Grinding stone rotations: 2500 rpm — Comparative Example 6 - Incision rate: 20 mm/min — Comparative Example 6 - Spark out time: 1 second — Comparative Example 6 - Grinding margin: 400 µm — Comparative Example 6 - Outer diameter difference between center and end part: 200 µm — Comparative Example 6 - Sulfur: 3 parts by mass — Comparative Example 21 - Sulfur: 0.2 parts by mass — Comparative Example 21 - Vulcanization temperature: 145° C. — Comparative Example 21 - Vulcanization temperature: 175° C. — Comparative Example 21 - Vulcanization temperature: 190° C. — Comparative Example 21 - Vulcanization temperature: 210° C. — Comparative Example 21 - UV integrated amount: 3000 mJ/cm² — Comparative Example 21

Relevant Experimental Evidence - Example 1: Showed preparation of an NBR-containing unvulcanized rubber composition, extrusion molding, air vulcanization at 160°C for 1 hour, and UV

surface treatment at 254 nm. - Comparative Example 5: Demonstrated that omitting UV irradiation changes the evaluation baseline for the charging roller. - Comparative Example 6: Showed that grinding conditions and subsequent UV treatment can be varied, with explicit control of roller rotation, stone rotation, and grinding margin. - Comparative Example 21: Provided a broad formulation/process matrix showing the effects of different sulfur levels, vulcanization temperatures, and UV doses on roller performance.

Technical Relevance This patent is relevant to NBR because it uses NBR as a major rubber component in a functional vulcanized article and discloses a detailed compounding and curing route. While not a polymerization patent, it is useful for understanding how NBR formulation and cure conditions influence surface and electrical performance in engineered rubber parts.

3. CROSS-PATENT COMPARISON & SYNTHESIS TRENDS

- EP2507313B1 is centered on carboxylated acrylonitrile-butadiene latex formulation for glove films, whereas US10317811B2 uses NBR as a compounded rubber in a charging roller; the former is latex-dipping oriented and the latter is extrusion/vulcanization oriented.
- EP2507313B1 explicitly controls acrylonitrile/butadiene/carboxylic acid composition and latex solids/pH, while US10317811B2 emphasizes filler and cure-package loading (carbon black, ZnO, sulfur, TBzTD) rather than monomer composition.
- EP2507313B1 targets sulfur- and accelerator-free crosslinking using metal oxide and methacrylic-acid/self-crosslinking concepts; US10317811B2 uses conventional sulfur/TBzTD vulcanization followed by UV surface oxidation.
- Temperature profiles differ substantially: EP2507313B1 reports glove drying at 50-70°C and 80-120°C with curing at 150°C, while US10317811B2 molds at 100°C and vulcanizes at 160°C for 1 hour before UV irradiation.
- EP2507313B1 focuses on latex total solids of 18-30% and pH 9-10, whereas US10317811B2 reports a solid rubber compounding route with parts-by-mass formulation and process variables such as 107% take-up rate and 9000 mJ/cm² UV dose.

4. CONCLUSION

The evidence shows two distinct but complementary NBR-use pathways: accelerator-free carboxylated nitrile latex films for dipped gloves and conventionally vulcanized NBR compounds for functional rollers. For NBR synthesis and manufacturing, the most actionable parameters are acrylonitrile content in the 26-30 wt% range, controlled latex solids of 18-30%, pH 9-10, and carefully selected crosslinking chemistry; for compounded articles, the key variables are filler loading, sulfur/TBzTD cure package, vulcanization temperature, and post-cure surface treatment. Based on the patents, the best-supported synthesis/manufacturing strategy for NBR-related products is to tightly control monomer composition and then tailor cure chemistry to the end-use:

metal-oxide/self-crosslinking routes for accelerator-free gloves, and conventional sulfur-based vulcanization plus surface oxidation for engineered roller applications.

5. REFERENCES

- EP2507313B1 | Elastomeric rubber and rubber products without the use of vulcanizing ... | Not disclosed | EP | 2018 | <https://patents.google.com/patent/EP2507313B1/en>
- US10317811B2 | Charging member, method for producing same, process cartridge and ... | Not disclosed | US | 2019 | <https://patents.google.com/patent/US10317811B2/en>